RESTAURANT REVIEW ANALYSIS SYSTEM

ADITYA UNIVERSITY

SCHOOL OF ENGINEERING

Department of Computer Science and Engineering

DATA ANALYSIS ESSENTIALS

CERTIFICATE

This is to certify that the Cornerstone project work entitled "*RESTAURANT REVIEW ANALYSIS SYSTEM*" is submitted by

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in partial fulfilment of the requirements for the award of the B.Tech. degree in Computer Science and Engineering for the academic year 2026-2027.

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DECLARATION

We hereby declare that the project entitled **"RESTAURANT REVIEW ANALYSIS SYSTEM"** is our original work and is submitted to **ADITYA UNIVERSITY, Surampalem**, in partial fulfillment of the requirements for the award of the **B.Tech. degree in Computer Science and Engineering.**

We further declare that this project has not been submitted, either in full or in part, for the award of any degree or diploma in this or any other institution. We also confirm that this work has been carried out with academic honesty and complies with the **Institutional Academic Integrity and AI Usage Policy**.

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RESTAURANT REVIEW ANALYSIS SYSTEM

ABSTRACT

The rapid growth of online food delivery platforms and restaurant review websites has made it essential to analyze customer feedback effectively. This project, Restaurant Review Analysis System, is designed to collect, process, and analyze customer reviews to help users make informed dining decisions and assist restaurant owners in improving their services. The system utilizes Natural Language Processing (NLP) and sentiment analysis techniques to classify reviews as positive, negative, or neutral based on the opinions expressed by customers. It also identifies frequently mentioned aspects such as food quality, service, ambiance, cleanliness, pricing, and delivery experience. The proposed system provides an easy-to-use interface where customers can view restaurant ratings, submit reviews, and explore detailed sentiment summaries. Restaurant administrators can monitor customer satisfaction, track performance trends, and identify areas requiring improvement through analytical dashboards and reports. The system stores review history securely and generate visual insights to support better business decisions. By automating the review analysis process, the Restaurant Review Analysis System reduces manual effort, increases the accuracy of customer feedback interpretation, and enhances the overall user experience. This intelligent platform serves as a valuable tool for both customers and restaurant management by promoting transparency, improving service quality, and supporting data-driven decision-making in the hospitality industry.

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S.No.	Abbreviation	Full Form
1	DAE	Data Analysis Essentials
2	EDA	Exploratory Data Analysis
3	NLP	Natural Language Processing
4	CSV	Comma-Separated Values
5	XLSX	Microsoft Excel Spreadsheet
6	ID	Identification
7	AI	Artificial Intelligence
8	ML	Machine Learning

Table 1: Abbreviations

Chapter 1
Introduction

1.1 Introduction to the Project

The Restaurant Review Analysis System is a Python-based project for studying restaurant reviews and restaurant performance. The supplied presentation describes a workflow that reads a restaurant dataset containing restaurant names, locations, ratings and customer reviews, cleans the raw data, converts it into a structured form, and analyzes ratings and feedback using Pandas and NumPy.

The project focuses on making review information easier to inspect and interpret. It uses operations such as head(), tail(), shape, fillna(), sorting, conditional categorization, and groupby statistics. The presentation also includes visual outputs for average rating over time and rating distribution.

1.2 Problem Statement

Restaurant review data contains valuable information about customer satisfaction, but analysing raw data manually can be difficult and time-consuming. The dataset may contain missing values, unclean review text, and different rating values, making it difficult to compare restaurants and identify useful patterns. Therefore, the project aims to clean, analyse, and visualize restaurant review data using Pandas and NumPy to obtain meaningful insights into customer feedback and restaurant performance.

1.3 Objectives of the Project

- To import and understand the restaurant review dataset.
- To clean and preprocess the dataset using Pandas and NumPy.
- To handle missing values and duplicate records.
- To analyze the distribution of restaurant ratings.
- Comparing restaurants based on their overall ratings.
- To analyze customer reviews and feedback.
- To categorize restaurants based on their ratings.
- To visualize the analysis using bar charts, pie charts, and word-frequency graphs.

- To generate meaningful insights into customer satisfaction and restaurant performance.

1.4 Scope of the Project

The scope of the Restaurant Review Analysis System is to analyse restaurant reviews and ratings using Pandas and NumPy. The project includes importing the dataset from Kaggle into Google Colab, cleaning and preprocessing the data, and handling missing and duplicate values. It analyses food, service, ambience, and overall ratings to understand customer satisfaction. The system also compares restaurant-wise ratings and analyses customer feedback and review text. Bar charts, pie charts, and word-frequency graphs are used to present the results clearly. The project can be further enhanced by implementing review classification using NLP and machine learning techniques.

Chapter 2
Dataset Description and Understanding

2.1 Dataset Source

The presentation states that the dataset was downloaded from Kaggle.com, uploaded to Google Drive, and then accessed from Google Colab after mounting the drive. The code shown in the presentation reads the Excel file from /content/drive/MyDrive/dae_dataset.xlsx using sheet_name="Sheet1".

2.2 Dataset Description

The dataset is stored in an Excel file and contains 10 restaurant review records. Each record represents one customer review. The displayed shape is (10, 8), so the dataset contains 10 rows and 8 columns.

CODE:

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print(df)
```

OUTPUT:

```
  Review_ID  Restaurant_Name  Food_Rating  Service_Rating  Ambience_Rating  \
0      R001     Spice Garden            5             4.0              5.0
1      R002         Food Hub            3             NaN              3.0
2      R003      Tasty Bites            2             2.0              3.0
3      R004       Urban Cafe            4             5.0              NaN
4      R005       Green Leaf            3             3.0              4.0
5      R006    Royal Kitchen            5             5.0              5.0
6      R007      Food Corner            1             2.0              2.0
7      R008  The Grill House            4             4.0              NaN
8      R009     Cafe Delight            2             3.0              3.0
9      R010      Spice Villa            4             NaN              5.0

   Overall_Rating                        Review_Text  Feedback
0               5     Excellent food and great ambience  Positive
1               3    Food was okay but service was slow   Neutral
2               2                                   NaN  Negative
3               4         Good food and friendly staff  Positive
4               3       Nice place but food was average       NaN
5               5    Amazing food and excellent service  Positive
6               1                 Very poor experience       NaN
7               4  Delicious food and reasonable prices  Positive
8               2        Food quality needs improvement  Negative
9               4         Great ambience and tasty food  Positive
```

Fig. 1: Dataset loaded from the Excel file in Google Colab

2.3 Features / Attributes

S.No.	Attribute	Meaning
1	Review_ID	Unique identification of each review
2	Restaurant_Name	Name of the restaurant
3	Food_Rating	Rating given for food quality
4	Service_Rating	Rating given for service quality
5	Ambience_Rating	Rating given for ambience
6	Overall_Rating	Overall customer rating
7	Review_Text	Written customer review
8	Feedback	Customer feedback classification

Table 2: Dataset attributes and meanings

2.4 Target Variable, if applicable

The presentation identifies Feedback as the target variable. It classifies reviews into Positive, Neutral or Negative categories.

2.5 Dataset Statistics / Initial Exploration

S.No.	Statistic	Value
1	Number of Records	10
2	Number of Attributes	8
3	Minimum Overall Rating	1
4	Maximum Overall Rating	5
5	Average Overall Rating	3.3
6	Rating Scale	1–5
7	Duplicate Records	0

Table 3: Initial dataset statistics reported

CODE:

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
df.head()
```

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
df.head()
```

	Review_ID	Restaurant_Name	Food_Rating	Service_Rating	Ambience_Rating	Overall_Rating	Review_Text	Feedback
0	R001	Spice Garden	5	4.0	5.0	5	Excellent food and great ambience	Positive
1	R002	Food Hub	3	NaN	3.0	3	Food was okay but service was slow	Neutral
2	R003	Tasty Bites	2	2.0	3.0	2	NaN	Negative
3	R004	Urban Cafe	4	5.0	NaN	4	Good food and friendly staff	Positive
4	R005	Green Leaf	3	3.0	4.0	3	Nice place but food was average	NaN

Fig. 2: First five records displayed using head().

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
df.tail()
```

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
df.tail()
```

	Review_ID	Restaurant_Name	Food_Rating	Service_Rating	Ambience_Rating	Overall_Rating	Review_Text	Feedback
5	R006	Royal Kitchen	5	5.0	5.0	5	Amazing food and excellent service	Positive
6	R007	Food Corner	1	2.0	2.0	1	Very poor experience	NaN
7	R008	The Grill House	4	4.0	NaN	4	Delicious food and reasonable prices	Positive
8	R009	Cafe Delight	2	3.0	3.0	2	Food quality needs improvement	Negative
9	R010	Spice Villa	4	NaN	5.0	4	Great ambience and tasty food	Positive

Fig. 3: Last five records displayed using tail().

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Shape:",df.shape)
```

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Shape:",df.shape)

Shape: (10, 8)
```

Fig. 4: Dataset shape output: (10, 8).

Chapter 3
Data Preprocessing

3.1 Data Collection / Loading

The documented collection process begins with downloading the dataset from Kaggle.com. The file is placed in Google Drive, Google Drive is mounted in Google Colab, and the Excel dataset is read into a Pandas Data Frame.

CODE:

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print(df)
```

```
   Review_ID  Restaurant_Name  Food_Rating  Service_Rating  Ambience_Rating  \
0      R001     Spice Garden            5             4.0              5.0
1      R002         Food Hub            3             NaN              3.0
2      R003      Tasty Bites            2             2.0              3.0
3      R004       Urban Cafe            4             5.0              NaN
4      R005       Green Leaf            3             3.0              4.0
5      R006    Royal Kitchen            5             5.0              5.0
6      R007      Food Corner            1             2.0              2.0
7      R008  The Grill House            4             4.0              NaN
8      R009     Cafe Delight            2             3.0              3.0
9      R010      Spice Villa            4             NaN              5.0

   Overall_Rating                          Review_Text  Feedback
0               5    Excellent food and great ambience  Positive
1               3   Food was okay but service was slow   Neutral
2               2                                  NaN  Negative
3               4        Good food and friendly staff  Positive
4               3      Nice place but food was average       NaN
5               5   Amazing food and excellent service  Positive
6               1               Very poor experience       NaN
7               4 Delicious food and reasonable prices  Positive
8               2       Food quality needs improvement  Negative
9               4        Great ambience and tasty food  Positive
```

Fig. 1: Dataset loaded from the Excel file in Google Colab

3.2 Handling Missing Values

Missing values in the dataset were identified using Pandas functions such as isnull().sum(). Missing values in numerical rating columns such as Service_Rating and Ambience_Rating were handled by replacing them with the mean value of the respective columns. Missing

Review_Text values were replaced with “No review provided”, while missing Feedback values were replaced with “Unknown”. The fillna() function was used to perform these operations, making the dataset complete and suitable for further analysis.

CODE:

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
df["Ambience_Rating"] = df["Ambience_Rating"].fillna("Not rated")
df["Service_Rating"] = df["Service_Rating"].fillna("Not rated")
df["Review_Text"] = df["Review_Text"].fillna("Not reviewed")
df["Feedback"] = df["Feedback"].fillna("No feedback")
print("Dataset after  filling missing values")
print(df)
```

```
Dataset after  filling missing values
  Review_ID  Restaurant_Name  Food_Rating Service_Rating Ambience_Rating  \
0      R001     Spice Garden            5            4.0             5.0
1      R002         Food Hub            3      Not rated             3.0
2      R003      Tasty Bites            2            2.0             3.0
3      R004       Urban Cafe            4            5.0       Not rated
4      R005       Green Leaf            3            3.0             4.0
5      R006    Royal Kitchen            5            5.0             5.0
6      R007      Food Corner            1            2.0             2.0
7      R008  The Grill House            4            4.0       Not rated
8      R009     Cafe Delight            2            3.0             3.0
9      R010      Spice Villa            4      Not rated             5.0

   Overall_Rating                          Review_Text     Feedback
0               5    Excellent food and great ambience     Positive
1               3   Food was okay but service was slow      Neutral
2               2                       Not reviewed     Negative
3               4        Good food and friendly staff     Positive
4               3     Nice place but food was average  No feedback
5               5   Amazing food and excellent service     Positive
6               1                  Very poor experience  No feedback
7               4 Delicious food and reasonable prices     Positive
8               2       Food quality needs improvement     Negative
9               4        Great ambience and tasty food     Positive
```

Fig. 5: Missing-value handling using fillna() for Review_Text.

.

3.3 Other Project-Specific Preprocessing

The project sorts the Overall_Rating column in both ascending and descending order to compare restaurant ratings. A new Category column is created using the apply() function with a lambda function. Restaurants with a rating of 5 are categorized as Most Popular, those with a rating of 4 as Popular, and ratings below 4 as Not Popular. This helps in organizing and comparing restaurants based on their overall ratings.

CODE:

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Restaurants sorted by ratings(Ascending)")
print(df.sort_values(by = "Overall_Rating"))
```

```
Restaurants sorted by ratings(Ascending)
  Review_ID  Restaurant_Name  Food_Rating  Service_Rating  Ambience_Rating  \
6      R007      Food Corner            1             2.0              2.0
2      R003      Tasty Bites            2             2.0              3.0
8      R009     Cafe Delight            2             3.0              3.0
4      R005       Green Leaf            3             3.0              4.0
1      R002         Food Hub            3             NaN              3.0
7      R008  The Grill House            4             4.0              NaN
9      R010      Spice Villa            4             NaN              5.0
3      R004       Urban Cafe            4             5.0              NaN
5      R006    Royal Kitchen            5             5.0              5.0
0      R001     Spice Garden            5             4.0              5.0

   Overall_Rating                          Review_Text  Feedback
6               1                Very poor experience       NaN
2               2                                NaN  Negative
8               2      Food quality needs improvement  Negative
4               3     Nice place but food was average       NaN
1               3   Food was okay but service was slow   Neutral
7               4  Delicious food and reasonable prices  Positive
9               4       Great ambience and tasty food  Positive
3               4        Good food and friendly staff  Positive
5               5   Amazing food and excellent service  Positive
0               5    Excellent food and great ambience  Positive
```

Fig. 6: Ascending-order sorting output.

CODE:

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Restaurants sorted by ratings(Descending)")
print(df.sort_values(by = "Overall_Rating",ascending = False))
```

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Restaurants sorted by ratings(Descending)")
print(df.sort_values(by = "Overall_Rating",ascending = False))

Restaurants sorted by ratings(Descending)
  Review_ID  Restaurant_Name  Food_Rating  Service_Rating  Ambience_Rating  \
0      R001     Spice Garden            5             4.0              5.0
5      R006    Royal Kitchen            5             5.0              5.0
9      R010      Spice Villa            4             NaN              5.0
3      R004       Urban Cafe            4             5.0              NaN
7      R008  The Grill House            4             4.0              NaN
1      R002         Food Hub            3             NaN              3.0
4      R005       Green Leaf            3             3.0              4.0
2      R003      Tasty Bites            2             2.0              3.0
8      R009     Cafe Delight            2             3.0              3.0
6      R007      Food Corner            1             2.0              2.0

   Overall_Rating                          Review_Text  Feedback
0               5    Excellent food and great ambience  Positive
5               5   Amazing food and excellent service  Positive
9               4        Great ambience and tasty food  Positive
3               4         Good food and friendly staff  Positive
7               4 Delicious food and reasonable prices  Positive
1               3   Food was okay but service was slow   Neutral
4               3      Nice place but food was average       NaN
2               2                                NaN  Negative
8               2       Food quality needs improvement  Negative
6               1                 Very poor experience       NaN
```

Fig. 7: Descending-order sorting output.

CODE:

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Printing After adding a new column")
df['Category'] = df['Overall_Rating'].apply(
    lambda x:'Most Popular' if x==5
    else 'Popular' if x==4
    else 'Not Popular')
print(df)
```

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Printing After adding a new column")
df['Category'] = df['Overall_Rating'].apply(
      lambda x:'Most Popular' if x==5
      else 'Popular' if x==4
      else 'Not Popular')
print(df)
```

```
Printing After adding a new column
  Review_ID  Restaurant_Name  Food_Rating  Service_Rating  Ambience_Rating  \
0      R001     Spice Garden            5             4.0              5.0
1      R002         Food Hub            3             NaN              3.0
2      R003      Tasty Bites            2             2.0              3.0
3      R004       Urban Cafe            4             5.0              NaN
4      R005       Green Leaf            3             3.0              4.0
5      R006    Royal Kitchen            5             5.0              5.0
6      R007      Food Corner            1             2.0              2.0
7      R008  The Grill House            4             4.0              NaN
8      R009     Cafe Delight            2             3.0              3.0
9      R010      Spice Villa            4             NaN              5.0

   Overall_Rating                          Review_Text  Feedback  \
0               5     Excellent food and great ambience  Positive
1               3    Food was okay but service was slow   Neutral
2               2                                   NaN  Negative
3               4          Good food and friendly staff  Positive
4               3       Nice place but food was average       NaN
5               5    Amazing food and excellent service  Positive
6               1                Very poor experience       NaN
7               4  Delicious food and reasonable prices  Positive
8               2        Food quality needs improvement  Negative
9               4         Great ambience and tasty food  Positive
```

	Category
0	Most Popular
1	Not Popular
2	Not Popular
3	Popular
4	Not Popular
5	Most Popular
6	Not Popular
7	Popular
8	Not Popular
9	Popular

Fig. 8: Rating-based category creation using apply() and lambda conditions.

Chapter 4

System Design and Methodology

4.1 Proposed Methodology

The methodology follows the sequence documented in the presentation: acquire the Excel dataset, connect it through Google Drive and Google Colab, inspect the data, handle missing review text, sort and categorize ratings, calculate feedback-wise statistics, and visualize rating patterns. The workflow below is derived only from these documented operations.

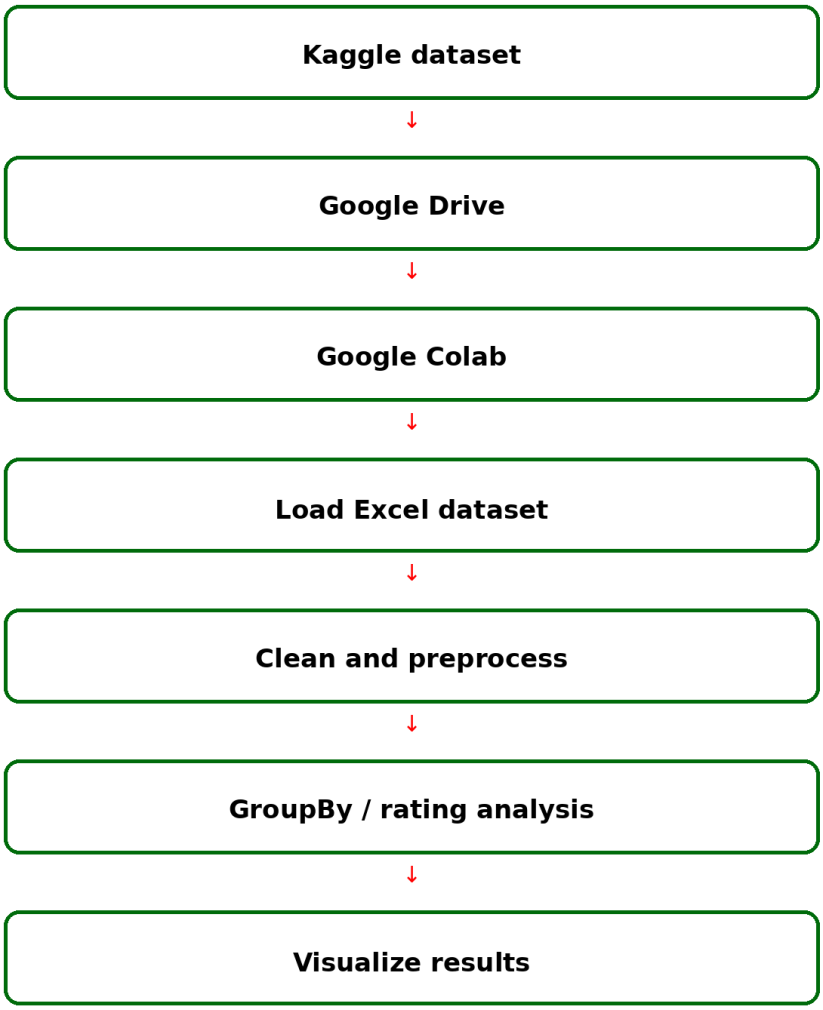


Fig. 9: Project workflow derived from the documented presentation process.

4.2 System Workflow / Architecture

The system is organized around a Data Frame-based analysis pipeline. The input is the Excel dataset. Pandas provides loading, inspection, missing-value handling, sorting, conditional transformation and groupby operations. NumPy is identified in the presentation as a suitable numerical analysis library. Google Colab provides the execution environment, and Google Drive provides access to the stored file.

4.3 Flowchart

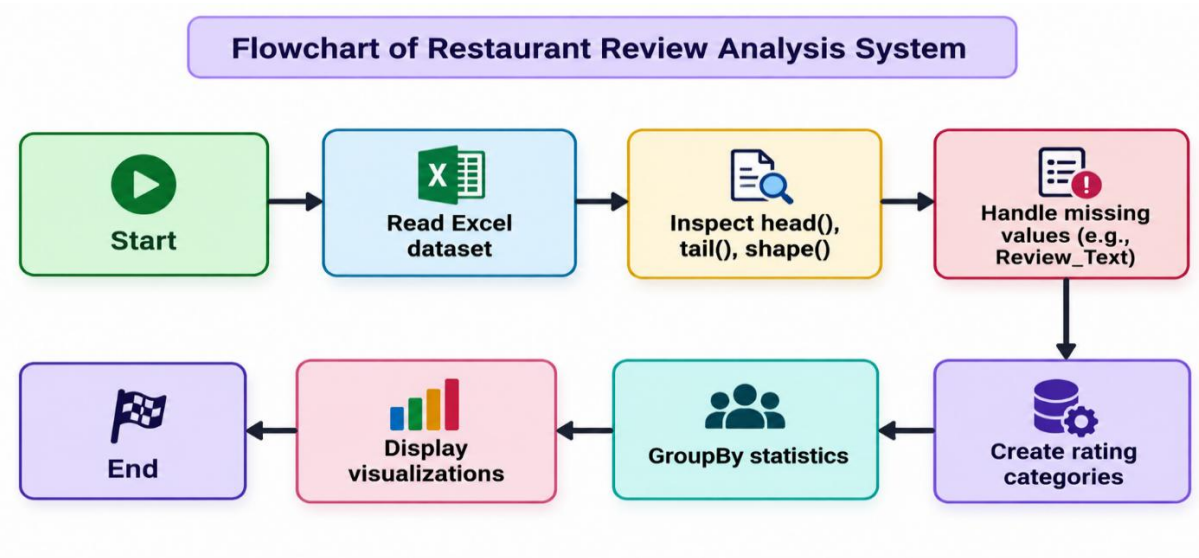


Fig. 10: Flowchart derived from the documented preprocessing and analysis steps.

4.4 Tools and Technologies Used

S.No.	Tool / Technology	Purpose
1	Kaggle	Dataset source
2	Google Drive	Dataset storage
3	Google Colab	Development environment
4	Python	Programming language
5	Pandas	Data cleaning and analysis
6	NumPy	Numerical operations
7	Matplotlib	Data visualization
8	Word Frequency Analysis	Review text analysis

The Restaurant Review Analysis System uses various tools and technologies for data collection, processing, analysis, and visualization. The dataset was obtained from Kaggle and stored in Google Drive. Google Colab was used as the development environment, while Python was used for implementing the analysis. Pandas was used for importing, cleaning, organizing, and analysing the dataset, and NumPy was used for numerical calculations. Matplotlib was used to create visualizations such as bar charts and pie charts. These tools together helped in transforming the raw restaurant review data into meaningful insights.

4.5 Algorithm / Model Used

Project mainly uses descriptive data analysis techniques with Pandas and NumPy. Functions such as mean(), count(), max(), min(), groupby(), apply(), lambda, and sorting are used to analyse restaurant ratings, compare restaurants, and create rating categories. Prediction using statistical or machine-learning models is considered as a future enhancement of the project.

Chapter 5
Implementation

5.1 Development Environment

The presentation documents Google Colab as the development and execution environment, with Google Drive mounted to access the Excel dataset. The analysis is implemented with Python and Pandas, with NumPy identified as an analysis library.

5.2 Dataset Loading and Exploration

The implementation reads the Excel workbook Sheet1 and uses head(), tail() and shape to inspect the dataset. The shape output is (10, 8).

5.3 Preprocessing Implementation

The demonstrated preprocessing includes filling missing Review_Text values, sorting Overall_Rating, and creating a Category column with conditional rules. The presentation does not document scaling, standardization or encoding operations.

5.4 Exploratory Data Analysis (EDA)

The EDA uses feedback-wise aggregation. The project calculates the mean, count, maximum and minimum Overall_Rating for each Feedback category.

5.5 Model / Analysis Implementation

The main analysis is descriptive rather than predictive. The displayed code groups Overall_Rating by Feedback and computes summary statistics. These operations allow the project to compare the observed rating levels associated with Positive, Neutral and Negative feedback.

The groupby() function in Pandas is used to group data based on a particular column and perform calculations on each group. In the Restaurant Review Analysis System, groupby() is used to group restaurants based on their names and calculate their average overall ratings. This helps compare the performance of different restaurants and understand customer satisfaction.

CODE:

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
```

```
print("Finding the average of feedbacks based on rating")
avg_feedback = df.groupby("Feedback")["Overall_Rating"].mean()
print(avg_feedback)
print("Finding the count of feedbacks based on rating")
feeback_count = df.groupby("Feedback")["Overall_Rating"].count()
print(feeback_count)
print("Finding the maximum of feedbacks based on rating")
max_feedback = df.groupby("Feedback")["Overall_Rating"].max()
print(max_feedback)
print("Finding the minimum of feedbacks based on rating")
min_feedback = df.groupby("Feedback")["Overall_Rating"].min()
print(min_feedback)
```

```
Finding the average of feedbacks based on rating
Feedback
Negative    2.0
Neutral     3.0
Positive    4.4
Name: Overall_Rating, dtype: float64
Finding the count of feedbacks based on rating
Feedback
Negative    2
Neutral     1
Positive    5
Name: Overall_Rating, dtype: int64
Finding the maximum of feedbacks based on rating
Feedback
Negative    2
Neutral     3
Positive    5
Name: Overall_Rating, dtype: int64
Finding the minimum of feedbacks based on rating
Feedback
Negative    2
Neutral     3
Positive    4
Name: Overall_Rating, dtype: int64
```

Fig 11: Performing operations using groupby()

Table 5: Feedback-wise rating statistics

S.No.	Feedback	Number of Reviews	Average Overall Rating
1	Positive	5	4.4
2	Neutral	1	3.0
3	Negative	2	2.0
4	Unknown	2	2.0

5.6 Important Code Explanation

The following code operations are important in the Restaurant Review Analysis System:

- pd.read_excel() – Imports the restaurant review dataset from the Excel file.
- head() – Displays the first few records of the dataset.
- tail() – Displays the last few records of the dataset.
- shape – Returns the number of rows and columns in the dataset.
- isnull().sum() – Identifies the number of missing values in each column.
- fillna() – Handles missing values by replacing them with suitable values.
- sort_values() – Sorts the Overall_Rating column in ascending or descending order.
- apply() – Applies a function to each value in a selected column.
- lambda – Defines conditions for creating the restaurant rating categories.
- groupby() – Groups restaurants and calculates statistics such as average ratings.
- count() – Counts the number of available values.
- min() – Finds the minimum rating.
- max() – Finds the maximum rating.
- mean() – Calculates the average rating.
- value_counts() – Counts the occurrences of different rating or feedback categories.

Chapter 6
Data Visualization and Results

6.1 Bar Graph

The bar chart shows the ambience ratings of different restaurants in the dataset. The restaurant names are represented on the X-axis, while the ambience rating from 0 to 5 is represented on the Y-axis. The groupby() function is used to group the data according to restaurant name, and the sum() function calculates the total ambience rating for each restaurant. The chart shows that Spice Garden, Royal Kitchen, and Spice Villa have the highest ambience rating of 5, while Food Corner has the lowest rating of 2. The graph provides a clear comparison of customer ratings for the ambience of different restaurants.

CODE:

```
import matplotlib.pyplot as plt
restaurant_rating = df.groupby('Restaurant_Name')['Ambience_Rating'].sum()
plt.figure(figsize=(7,5))
plt.bar(restaurant_rating.index, restaurant_rating.values,color = "steelblue")
plt.xlabel('Restaurant Name')
plt.ylabel('Ambience Rating')
plt.title('Ambience Rating by Restaurant')
plt.tight_layout()
plt.savefig("bar_graph.png",dpi = 120)
plt.close()
```

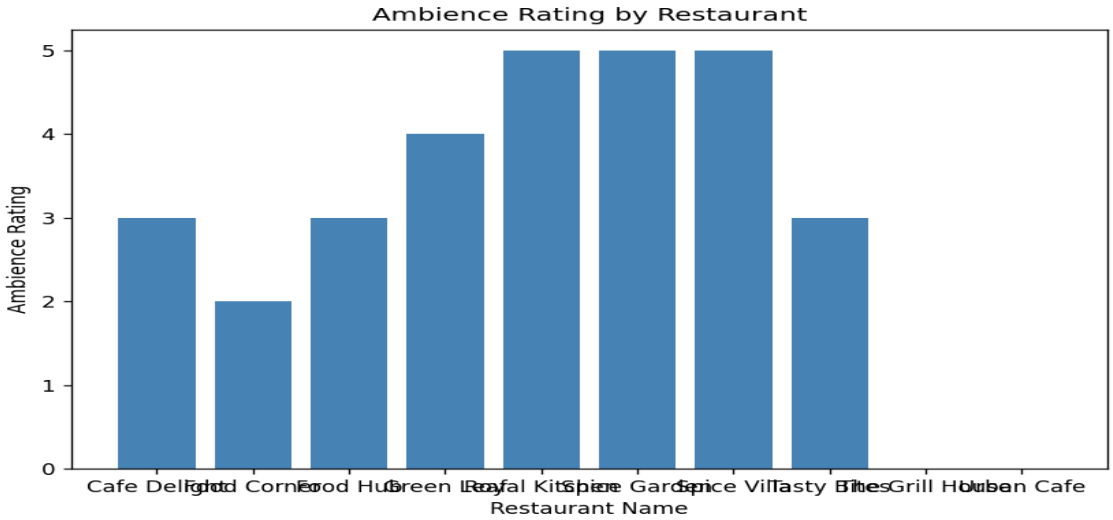


Fig 12: Bar Chart on Ambience Rating by Department

6.2 Pie Chart

The pie chart represents the distribution of customer feedback in the restaurant review dataset. The value_counts() function is used to count the number of reviews in each feedback category. The chart shows that Positive feedback accounts for 62.5%, Negative feedback accounts for 25%, and Neutral feedback accounts for 12.5%. This indicates that positive reviews form the largest portion of the available feedback, followed by negative and neutral reviews. The pie chart provides a clear visual representation of the overall customer feedback distribution.

CODE:

```
feedback_count = df['Feedback'].value_counts()

plt.figure(figsize=(6,6))

plt.pie(feedback_count.values,labels = feedback_count.index, autopct = "%1.1f%%", startangle = 90)

plt.title('Feedback Distribution')

plt.tight_layout()

plt.savefig("pie_chart.png",dpi = 120)

plt.close()
```

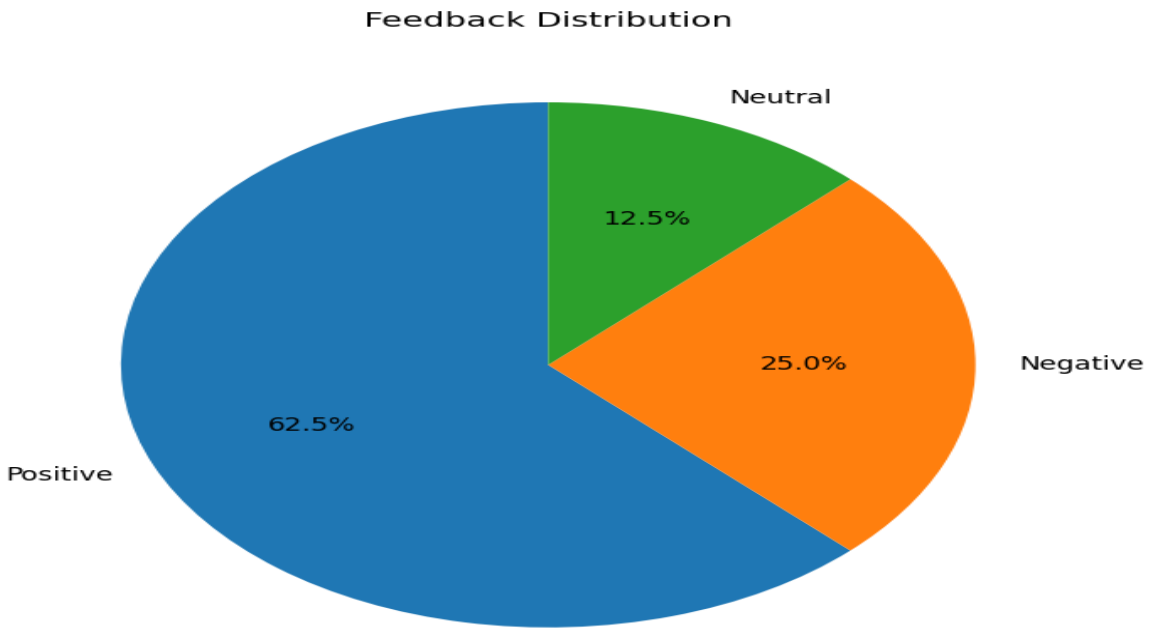


Fig 13: Pie Chart on Feedback Distribution

6.3 Box Plot

The box plot shows the distribution of overall ratings for different feedback categories. The Positive feedback category has ratings mainly between 4 and 5, indicating higher customer satisfaction. Neutral feedback is centered around a rating of 3, while Negative feedback is centered around a rating of 2. The plot helps compare the rating patterns across different feedback categories and shows the relationship between customer feedback and overall ratings.

CODE:

```
feedback_values = df["Feedback"].unique()

data_by_feedback = [df[df["Feedback"] == feedback]["Overall_Rating"].dropna() for feedback in feedback_values]

plt.figure(figsize=(8,5))

plt.boxplot(data_by_feedback, tick_labels = feedback_values)

plt.xlabel("Feedback")

plt.ylabel("Overall Rating")

plt.title("Overall Rating Distribution by Feedback")

plt.tight_layout()

plt.savefig("box_plot.png",dpi = 120)

plt.close()
```

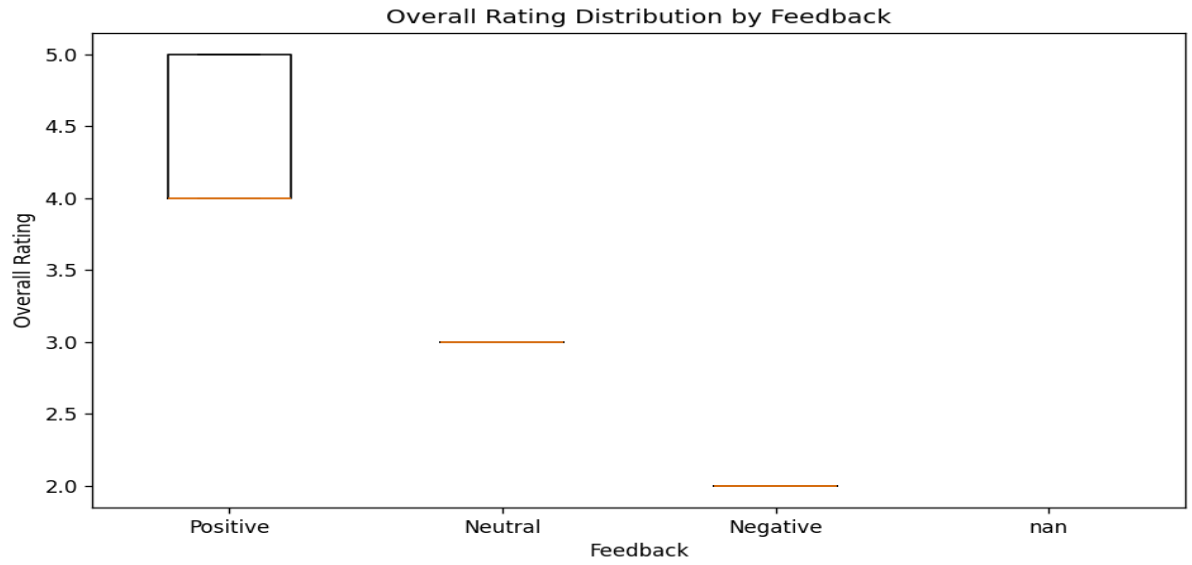


Fig 14: Box Plot on Overall Rating Distribution by Feedback

6.4 Histogram

The histogram shows the distribution of food ratings given by customers. The Food_Rating values range from 1 to 5, with the highest frequency observed for a rating of 4. A rating of 1 occurs least frequently, while ratings of 2, 3, and 5 occur with moderate frequency. The histogram helps visualize how customer ratings for food quality are distributed across the restaurants and provides an overview of customer satisfaction with the food.

CODE:

```
plt.figure(figsize=(7,5))

plt.hist(df["Food_Rating"].dropna(), bins = 8, color = "mediumseagreen", edgecolor = "black")

plt.xlabel("Food_Rating")

plt.ylabel("Frequency")

plt.title("Distribution of food ratings")

plt.tight_layout()

plt.savefig("histogram.png",dpi = 120)

plt.close()
```

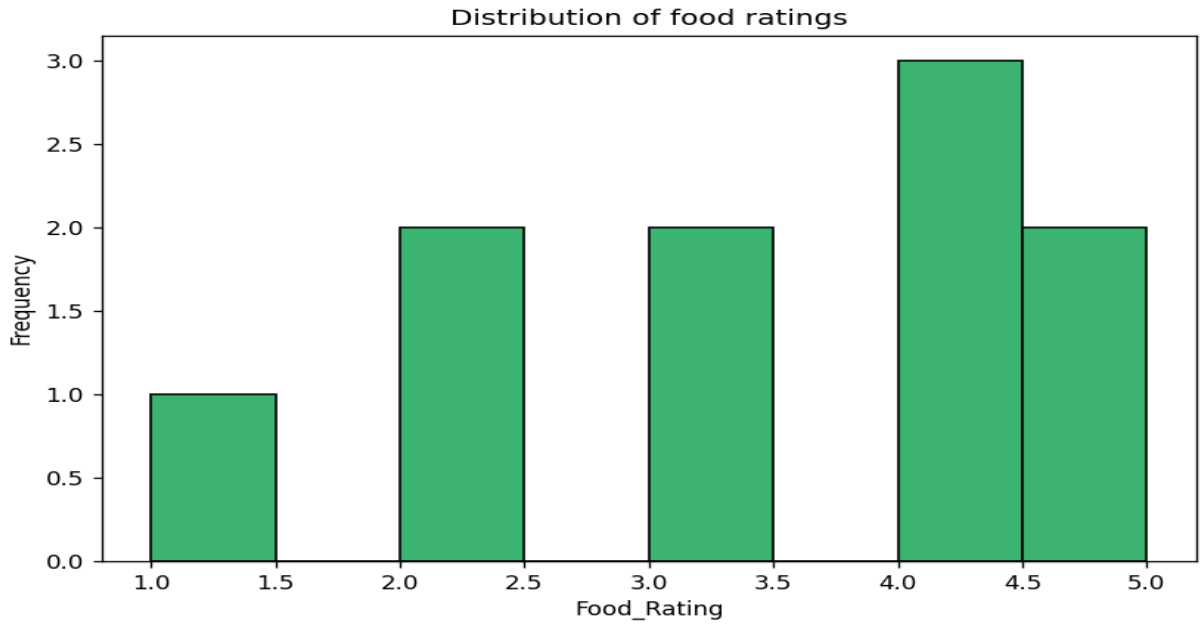


Fig 15: Histogram on Distribution of food ratings

6.5 Line Chart

The graph consists of two line charts that show the variation in food and service ratings across different records. The first chart represents the Food Rating Trend, where the ratings vary between 1 and 5, with higher ratings indicating greater customer satisfaction with food quality. The second chart represents the Service Rating Trend, where service ratings also range from 2 to 5. The graphs help compare the changes in food and service ratings across the records and identify variations in customer satisfaction.

CODE:

```
fig, axes = plt.subplots(2,1,figsize=(8,8),sharex = True)

axes[0].plot(df.index,df["Food_Rating"],marker="o",color="royalblue",label="Food Rating")

axes[0].set_title("Food Rating Trend")

axes[0].set_ylabel("Food Rating")

axes[0].legend()

axes[0].grid(True,alpha=0.3)

axes[1].plot(df.index,df["Service_Rating"],marker="s",color="darkorange",label="Service Rating")

axes[1].set_title("Service Rating Trend")

axes[1].set_xlabel("Record Index")

axes[1].set_ylabel("Service Rating")

axes[1].legend()

axes[1].grid(True,alpha=0.3)

plt.tight_layout()

plt.savefig("line_chart_subplots.png",dpi = 120)

plt.close()
```

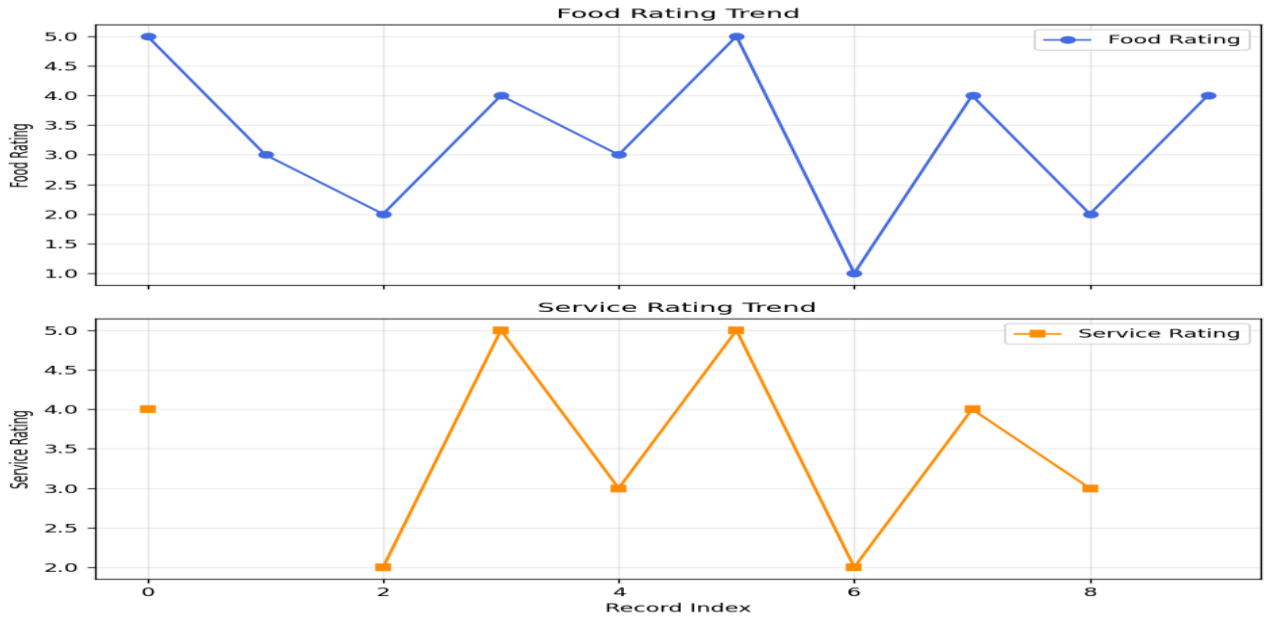


Fig 16: Line Chart on Food and Service Rating Trend

6.6 Scatter Plot

The scatter plot shows the relationship between Food_Rating and Overall_Rating. The Food_Rating is represented on the X-axis, while the Overall_Rating is represented on the Y-axis. Each point represents a restaurant record. The graph shows that higher food ratings are generally associated with higher overall ratings. For example, a food rating of 5 corresponds to an overall rating of 5, while a food rating of 1 corresponds to an overall rating of 1. This indicates a positive relationship between food quality and the overall customer rating in the dataset.

CODE:

```
plt.figure(figsize=(7,5))

plt.scatter(df["Food_Rating"],df["Overall_Rating"],c = "crimson", edgecolor = "black", alpha = 0.75)

plt.xlabel("Food Rating")

plt.ylabel("Overall Rating")

plt.title("Food Rating vs Overall Rating")

plt.tight_layout()

plt.savefig("scatter_plot.png",dpi = 120)

plt.close()
```

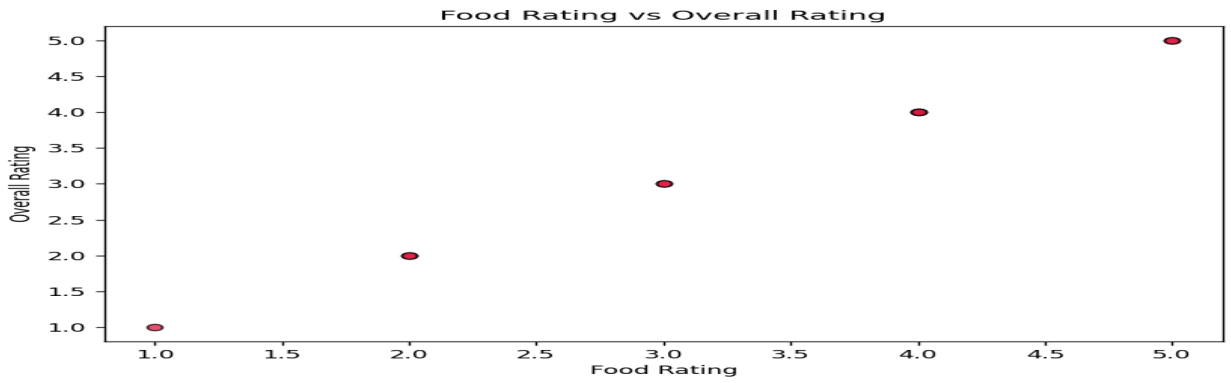


Fig 17: Scatter Plot on Food Rating vs Overall Rating

6.7 Project-Specific Visualizations

The Restaurant Review Analysis System uses different visualizations to present the results of the data analysis clearly. These visualizations help understand customer ratings, feedback patterns, and relationships between different rating attributes.

- Ambience Rating by Restaurant – compares the ambience ratings of different restaurants using a bar chart.
- Feedback Distribution – represents the proportion of Positive, Negative, and Neutral feedback using a pie chart.
- Overall Rating Distribution by Feedback – uses a box plot to compare overall ratings across different feedback categories.
- Food Rating Distribution – uses a histogram to show the frequency of different food ratings.
- Food and Service Rating Trends – uses line charts to show the variation in food and service ratings across records.
- Food Rating vs Overall Rating – uses a scatter plot to show the relationship between food ratings and overall ratings.

6.8 Interpretation of Visualizations

The visualizations provide a clear understanding of the restaurant review dataset and customer satisfaction patterns. The ambience rating bar chart compares the ambience ratings of different restaurants, while the feedback distribution pie chart shows the proportion of positive, negative, and neutral reviews. The box plot demonstrates that positive feedback is generally associated with higher overall ratings, whereas negative feedback is associated with lower ratings The scatter plot indicates a positive relationship between Food_Rating and Overall_Rating, where higher food ratings are generally associated with higher overall ratings. Overall, these visualizations make the dataset easier to understand and help identify customer satisfaction patterns.

Chapter 7
Model Evaluation / Analysis of Results

7.1 Evaluation Metrics

The completed presentation does not report predictive-model evaluation metrics such as accuracy, precision, recall or F1-score. The documented evaluation is descriptive and uses mean, count, maximum and minimum Overall_Rating grouped by Feedback.

7.2 Experimental Results

Table 6: Feedback-wise evaluation summary.

S.No.	Feedback	Rating Range	Average Rating	Evaluation
1	Positive	4–5	4.4	Higher satisfaction
2	Neutral	3	3.0	Moderate satisfaction
3	Negative	1–2	2.0	Lower satisfaction

This table presents the evaluation of restaurant reviews based on customer feedback and overall ratings. Positive feedback has a rating range of 4–5, indicating higher customer satisfaction, while neutral feedback has a rating of 3, representing moderate satisfaction. Negative feedback falls within the 1–2 rating range, indicating lower customer satisfaction. The table helps in understanding the relationship between customer feedback and overall restaurant ratings

7.3 Result Interpretation

The descriptive results show different observed rating levels across the three non-empty Feedback categories. Positive feedback has a displayed mean of 4.4, while Neutral and Negative feedback have displayed means of 3.0 and 2.0 respectively. The maximum and minimum values also show the range present within each group.

7.4 Discussion

The project demonstrates a complete basic analysis pipeline from Excel data loading to descriptive statistics and visualization. The presentation does not provide a predictive model, accuracy measurement or a validated train/test evaluation. Therefore, conclusions are limited to the displayed descriptive analysis and visual outputs.

Chapter 8

Output Screenshots

8.1 Input / Dataset Screenshots

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print(df)
```

```
  Review_ID  Restaurant_Name  Food_Rating  Service_Rating  Ambience_Rating  \
0      R001     Spice Garden            5             4.0              5.0
1      R002         Food Hub            3             NaN              3.0
2      R003      Tasty Bites            2             2.0              3.0
3      R004       Urban Cafe            4             5.0              NaN
4      R005       Green Leaf            3             3.0              4.0
5      R006    Royal Kitchen            5             5.0              5.0
6      R007      Food Corner            1             2.0              2.0
7      R008  The Grill House            4             4.0              NaN
8      R009     Cafe Delight            2             3.0              3.0
9      R010      Spice Villa            4             NaN              5.0

   Overall_Rating                        Review_Text  Feedback
0               5    Excellent food and great ambience  Positive
1               3   Food was okay but service was slow   Neutral
2               2                                NaN  Negative
3               4        Good food and friendly staff  Positive
4               3      Nice place but food was average       NaN
5               5   Amazing food and excellent service  Positive
6               1              Very poor experience       NaN
7               4 Delicious food and reasonable prices  Positive
8               2       Food quality needs improvement  Negative
9               4        Great ambience and tasty food  Positive
```

Fig. 1: Dataset input and displayed records in Google Colab.

8.2 Preprocessing Outputs

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
df.head()
```

	Review_ID	Restaurant_Name	Food_Rating	Service_Rating	Ambience_Rating	Overall_Rating	Review_Text	Feedback
0	R001	Spice Garden	5	4.0	5.0	5	Excellent food and great ambience	Positive
1	R002	Food Hub	3	NaN	3.0	3	Food was okay but service was slow	Neutral
2	R003	Tasty Bites	2	2.0	3.0	2	NaN	Negative
3	R004	Urban Cafe	4	5.0	NaN	4	Good food and friendly staff	Positive
4	R005	Green Leaf	3	3.0	4.0	3	Nice place but food was average	NaN

Fig. 2: head() output.

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
df.tail()
```

	Review_ID	Restaurant_Name	Food_Rating	Service_Rating	Ambience_Rating	Overall_Rating	Review_Text	Feedback
5	R006	Royal Kitchen	5	5.0	5.0	5	Amazing food and excellent service	Positive
6	R007	Food Corner	1	2.0	2.0	1	Very poor experience	NaN
7	R008	The Grill House	4	4.0	NaN	4	Delicious food and reasonable prices	Positive
8	R009	Cafe Delight	2	3.0	3.0	2	Food quality needs improvement	Negative
9	R010	Spice Villa	4	NaN	5.0	4	Great ambience and tasty food	Positive

Fig. 3: tail() output.

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Shape:",df.shape)
```

```
Shape: (10, 8)
```

Fig. 4: shape output.

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
df["Ambience_Rating"] = df["Ambience_Rating"].fillna("Not rated")
df["Service_Rating"] = df["Service_Rating"].fillna("Not rated")
df["Review_Text"] = df["Review_Text"].fillna("Not reviewed")
df["Feedback"] = df["Feedback"].fillna("No feedback")
print("Dataset after  filling missing values")
print(df)
```

```
Dataset after  filling missing values
  Review_ID  Restaurant_Name  Food_Rating Service_Rating Ambience_Rating  \
0      R001     Spice Garden            5            4.0             5.0
1      R002         Food Hub            3      Not rated             3.0
2      R003      Tasty Bites            2            2.0             3.0
3      R004       Urban Cafe            4            5.0       Not rated
4      R005       Green Leaf            3            3.0             4.0
5      R006    Royal Kitchen            5            5.0             5.0
6      R007      Food Corner            1            2.0             2.0
7      R008  The Grill House            4            4.0       Not rated
8      R009     Cafe Delight            2            3.0             3.0
9      R010      Spice Villa            4      Not rated             5.0

   Overall_Rating                          Review_Text     Feedback
0               5    Excellent food and great ambience     Positive
1               3   Food was okay but service was slow      Neutral
2               2                       Not reviewed     Negative
3               4        Good food and friendly staff     Positive
4               3     Nice place but food was average  No feedback
5               5   Amazing food and excellent service     Positive
6               1                 Very poor experience  No feedback
7               4 Delicious food and reasonable prices     Positive
8               2       Food quality needs improvement     Negative
9               4        Great ambience and tasty food     Positive
```

Fig. 5: fillna() preprocessing output.

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Restaurants sorted by ratings(Ascending)")
print(df.sort_values(by = "Overall_Rating"))
```

```
Restaurants sorted by ratings(Ascending)
  Review_ID  Restaurant_Name  Food_Rating  Service_Rating  Ambience_Rating  \
6      R007      Food Corner            1             2.0              2.0
2      R003      Tasty Bites            2             2.0              3.0
8      R009     Cafe Delight            2             3.0              3.0
4      R005       Green Leaf            3             3.0              4.0
1      R002         Food Hub            3             NaN              3.0
7      R008  The Grill House            4             4.0              NaN
9      R010      Spice Villa            4             NaN              5.0
3      R004       Urban Cafe            4             5.0              NaN
5      R006    Royal Kitchen            5             5.0              5.0
0      R001     Spice Garden            5             4.0              5.0

   Overall_Rating                          Review_Text  Feedback
6               1                Very poor experience       NaN
2               2                                NaN  Negative
8               2       Food quality needs improvement  Negative
4               3      Nice place but food was average       NaN
1               3   Food was okay but service was slow   Neutral
7               4  Delicious food and reasonable prices  Positive
9               4        Great ambience and tasty food  Positive
3               4         Good food and friendly staff  Positive
5               5   Amazing food and excellent service  Positive
0               5    Excellent food and great ambience  Positive
```

Fig. 6: ascending sorting output.

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Restaurants sorted by ratings(Descending)")
print(df.sort_values(by = "Overall_Rating",ascending = False))
```

```
Restaurants sorted by ratings(Descending)
  Review_ID  Restaurant_Name  Food_Rating  Service_Rating  Ambience_Rating  \
0      R001     Spice Garden            5             4.0              5.0
5      R006    Royal Kitchen            5             5.0              5.0
9      R010      Spice Villa            4             NaN              5.0
3      R004       Urban Cafe            4             5.0              NaN
7      R008  The Grill House            4             4.0              NaN
1      R002         Food Hub            3             NaN              3.0
4      R005       Green Leaf            3             3.0              4.0
2      R003      Tasty Bites            2             2.0              3.0
8      R009     Cafe Delight            2             3.0              3.0
6      R007      Food Corner            1             2.0              2.0

   Overall_Rating                          Review_Text  Feedback
0               5    Excellent food and great ambience  Positive
5               5   Amazing food and excellent service  Positive
9               4        Great ambience and tasty food  Positive
3               4         Good food and friendly staff  Positive
7               4  Delicious food and reasonable prices  Positive
1               3   Food was okay but service was slow   Neutral
4               3      Nice place but food was average       NaN
2               2                                NaN  Negative
8               2       Food quality needs improvement  Negative
6               1                Very poor experience       NaN
```

Fig. 7: descending sorting output.

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Printing After adding a new column")
df['Category'] = df['Overall_Rating'].apply(
      lambda x:'Most Popular' if x==5
      else 'Popular' if x==4
      else 'Not Popular')
print(df)
```

```
Printing After adding a new column
  Review_ID  Restaurant_Name  Food_Rating  Service_Rating  Ambience_Rating  \
0      R001     Spice Garden            5             4.0              5.0
1      R002         Food Hub            3             NaN              3.0
2      R003      Tasty Bites            2             2.0              3.0
3      R004       Urban Cafe            4             5.0              NaN
4      R005       Green Leaf            3             3.0              4.0
5      R006    Royal Kitchen            5             5.0              5.0
6      R007      Food Corner            1             2.0              2.0
7      R008  The Grill House            4             4.0              NaN
8      R009     Cafe Delight            2             3.0              3.0
9      R010      Spice Villa            4             NaN              5.0

   Overall_Rating                         Review_Text  Feedback  \
0               5    Excellent food and great ambience  Positive
1               3   Food was okay but service was slow   Neutral
2               2                                  NaN  Negative
3               4        Good food and friendly staff  Positive
4               3      Nice place but food was average       NaN
5               5   Amazing food and excellent service  Positive
6               1               Very poor experience       NaN
7               4  Delicious food and reasonable prices  Positive
8               2       Food quality needs improvement  Negative
9               4        Great ambience and tasty food  Positive
```

	Category
0	Most Popular
1	Not Popular
2	Not Popular
3	Popular
4	Not Popular
5	Most Popular
6	Not Popular
7	Popular
8	Not Popular
9	Popular

Fig. 8: rating-category transformation.

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print(df.isnull().sum())
```

```
Review_ID          0
Restaurant_Name    0
Food_Rating        0
Service_Rating     2
Ambience_Rating    2
Overall_Rating     0
Review_Text        1
Feedback           2
dtype: int64
```

Fig. 18: isnull().sum() function

Setting Conditions: Printing the reviews whose overall rating is greater than 3

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
high_review = df[df['Overall_Rating']>=4]
print(high_review)
```

```
  Review_ID  Restaurant_Name  Food_Rating  Service_Rating  Ambience_Rating  \
0      R001     Spice Garden            5             4.0              5.0
3      R004       Urban Cafe            4             5.0              NaN
5      R006    Royal Kitchen            5             5.0              5.0
7      R008  The Grill House            4             4.0              NaN
9      R010      Spice Villa            4             NaN              5.0

   Overall_Rating                         Review_Text  Feedback
0               5     Excellent food and great ambience  Positive
3               4          Good food and friendly staff  Positive
5               5    Amazing food and excellent service  Positive
7               4  Delicious food and reasonable prices  Positive
9               4         Great ambience and tasty food  Positive
```

Fig. 19: Setting conditions

Printing the attributes of the dataset

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
df.columns
```

```
Index(['Review_ID', 'Restaurant_Name', 'Food_Rating', 'Service_Rating',
       'Ambience_Rating', 'Overall_Rating', 'Review_Text', 'Feedback'],
      dtype='object')
```

Fig. 20: Printing attributes

Performing the operations on the dataset using aggregate functions

```
import pandas as pd
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')
print("Finding the average of feedbacks based on rating")
avg_feedback = df.groupby("Feedback")["Overall_Rating"].mean()
print(avg_feedback)
print("Finding the count of feedbacks based on rating")
feeback_count = df.groupby("Feedback")["Overall_Rating"].count()
print(feeback_count)
print("Finding the maximum of feedbacks based on rating")
max_feedback = df.groupby("Feedback")["Overall_Rating"].max()
print(max_feedback)
print("Finding the minimum of feedbacks based on rating")
min_feedback = df.groupby("Feedback")["Overall_Rating"].min()
print(min_feedback)
```

```
Finding the average of feedbacks based on rating
Feedback
Negative    2.0
Neutral     3.0
Positive    4.4
Name: Overall_Rating, dtype: float64
Finding the count of feedbacks based on rating
Feedback
Negative    2
Neutral     1
Positive    5
Name: Overall_Rating, dtype: int64
Finding the maximum of feedbacks based on rating
Feedback
Negative    2
Neutral     3
Positive    5
Name: Overall_Rating, dtype: int64
Finding the minimum of feedbacks based on rating
Feedback
Negative    2
Neutral     3
Positive    4
Name: Overall_Rating, dtype: int64
```

Fig 11: Performing operations using groupby()

8.3 Visualization Outputs

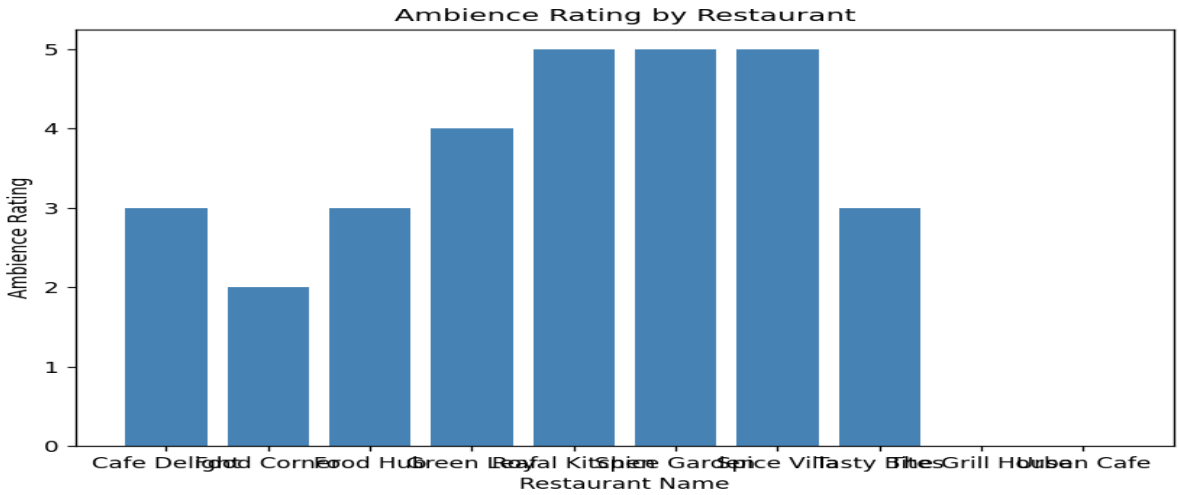


Fig. 12: Bar Chart

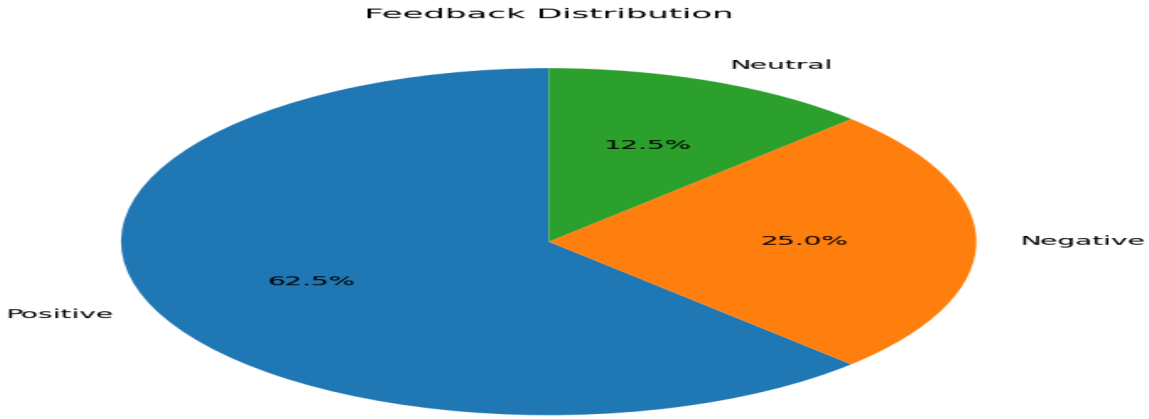


Fig. 13: Pie Chart

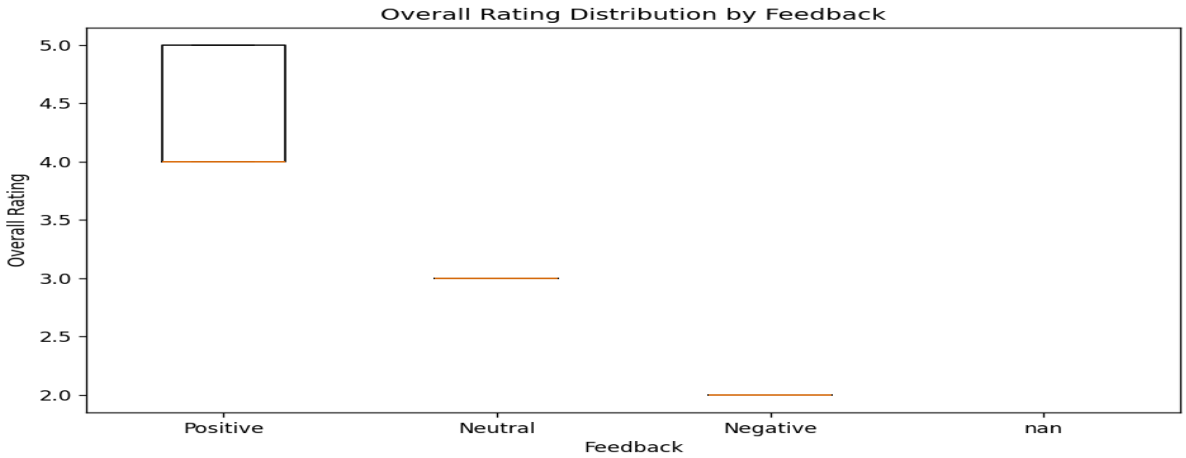


Fig. 14: Box Plot

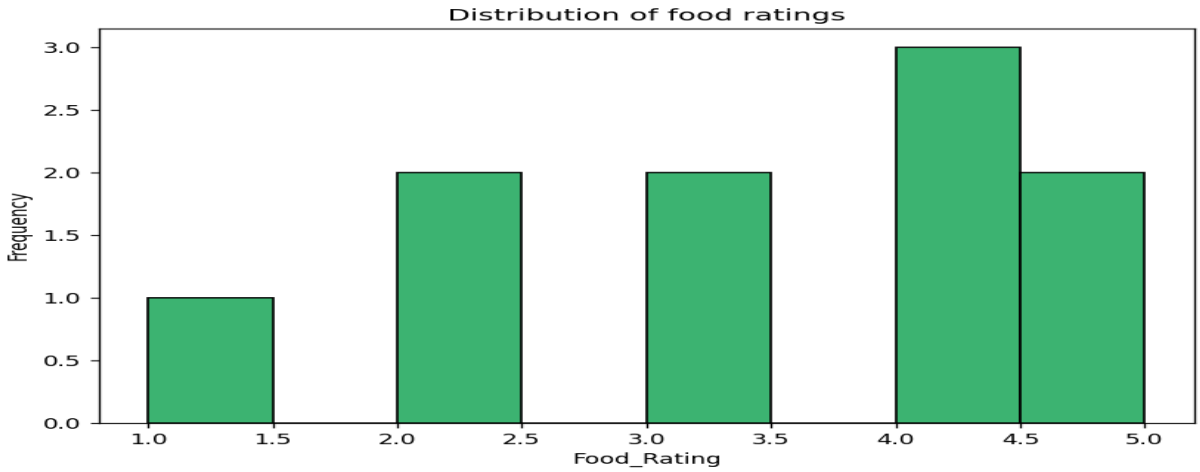


Fig. 15: Histogram

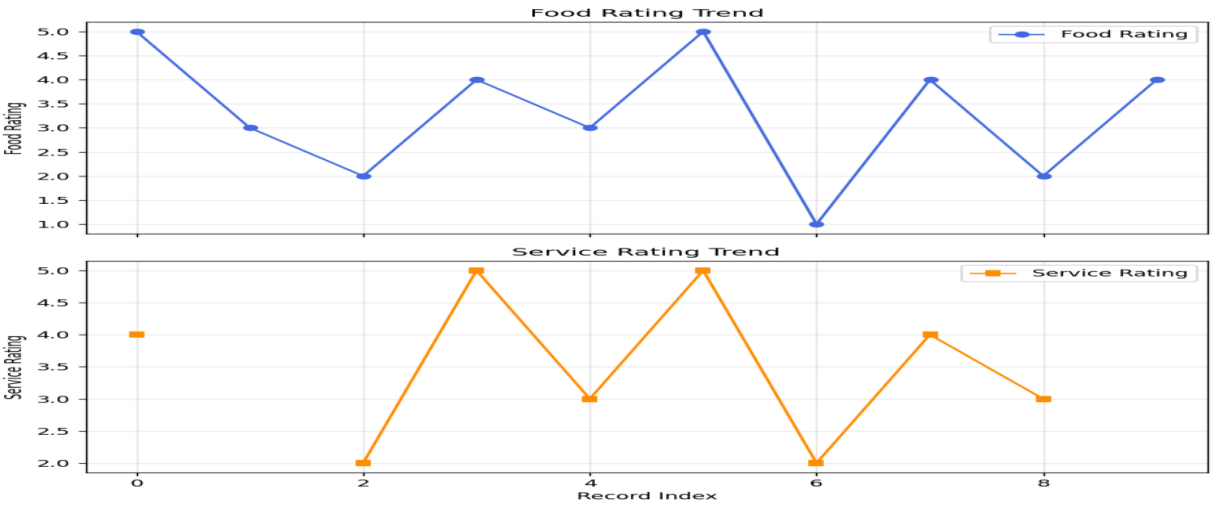


Fig. 16: Line Chart

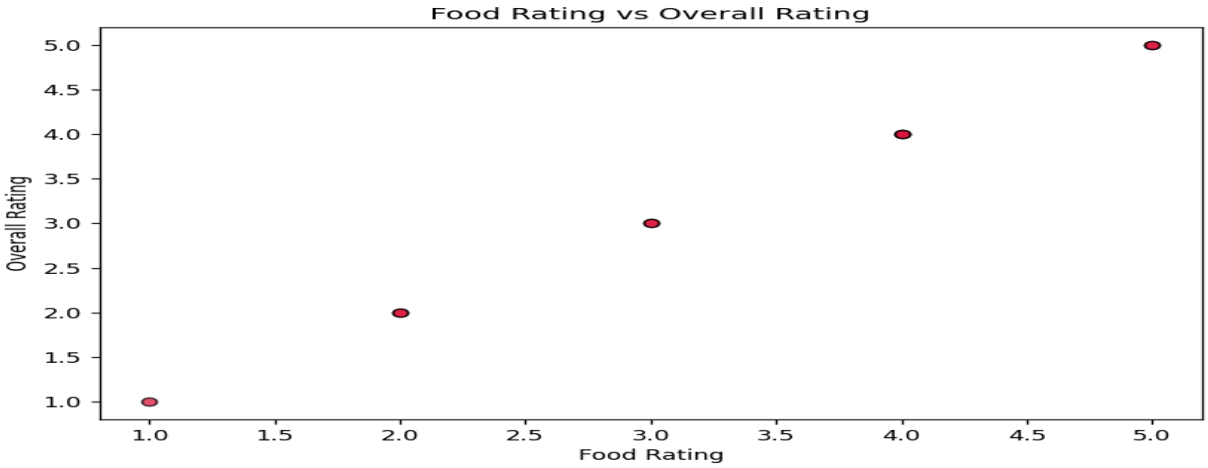


Fig. 17: Scatter Plot

Chapter 9
Conclusion and Future Enhancements

9.1 Conclusion

The Restaurant Review Analysis System demonstrates analysis of customer reviews and restaurant ratings using Python, Pandas and NumPy. The documented workflow loads an Excel dataset, handles missing review text, sorts and categorizes ratings, and calculates descriptive statistics using groupby operations. The supplied outputs provide feedback-wise rating summaries and visualizations for rating patterns.

9.2 Limitations

- The supplied presentation documents only 10 dataset records.
- No predictive machine-learning model or predictive evaluation metric is reported.
- No duplicate-removal, feature-scaling, standardization, label-encoding or one-hot-encoding step is documented.
- The rating-distribution visualization contains frequencies that do not match the 10-record dataset size, so those chart values are not used as dataset-level counts.
- The complete source-code file was not included with the supplied attachments.

9.3 Future Enhancements

- Extend the system with sentiment analysis to automatically classify reviews as positive, negative or neutral.
- Predict customer satisfaction using statistical or machine-learning models.
- Enable personalized restaurant recommendations based on customer reviews and ratings.
- Add real-time review data for continuous monitoring of customer feedback.
- Build restaurant dashboards for ongoing performance and rating analysis.
- Explore food quality, service, price and ambience factors as additional predictors of customer satisfaction.

10
References

- DAE Course Textbook and Study Material – Data Analysis Essentials.
- Fabio Nelli – Python Data Analytics, Apress.
- Wes McKinney – Python for Data Analysis, O'Reilly.
- Alberto Boschetti & Luca Massaron – Python Data Science Essentials, Packt.
- Kaggle – Restaurant Review Dataset.
- Pandas and NumPy Documentation.
- Matplotlib Documentation.

11

Appendix – Source Code

The following source code was used to import, clean, analyse, and visualize the Restaurant Review dataset using Python, Pandas, NumPy, and Matplotlib.

```
# Import libraries
import pandas as pd
import numpy as np

# Import the dataset
df = pd.read_excel('/content/drive/MyDrive/dae_dataset.xlsx',sheet_name = 'Sheet1')

# Display first and last records
print(df.head())
print(df.tail())

# Check dataset shape
print(df.shape)

# Check missing values
print(df.isnull().sum())

# Handle missing values
df["Ambience_Rating"] = df["Ambience_Rating"].fillna("Not rated")
df["Service_Rating"] = df["Service_Rating"].fillna("Not rated")
df["Review_Text"] = df["Review_Text"].fillna("Not reviewed")
df["Feedback"] = df["Feedback"].fillna("No feedback")

# Clean review text
df['Review_Text'] = df['Review_Text'].str.strip()
df['Review_Text'] = df['Review_Text'].str.lower()
```

```
# Rating statistics
avg_feedback = df.groupby("Feedback")["Overall_Rating"].mean()
feeback_count = df.groupby("Feedback")["Overall_Rating"].count()
max_feedback = df.groupby("Feedback")["Overall_Rating"].max()
min_feedback = df.groupby("Feedback")["Overall_Rating"].min()

# Sort ratings
print("Restaurants sorted by ratings(Ascending)")
print(df.sort_values(by = "Overall_Rating"))
print("Restaurants sorted by ratings(Descending)")
print(df.sort_values(by = "Overall_Rating",ascending = False))

# Create restaurant category
df['Category'] = df['Overall_Rating'].apply(
    lambda x:'Most Popular' if x==5
    else 'Popular' if x==4
    else 'Not Popular')

# Setting Conditions
high_review = df[df['Overall_Rating']>=4]
```